

Assignment-1

Student Details

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Division (Div): A

Subject Code: 701

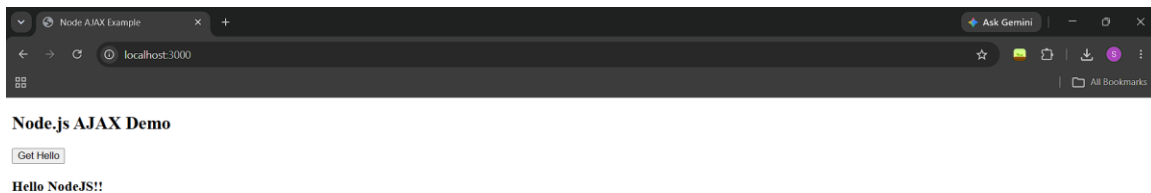
Problem 1

Problem Statement:

Develop nodejs application with following requirements:

- Develop a route "/gethello" with GET method. It displays "Hello NodeJS!!" as response.
- Make an HTML page and display.
- Call "/gethello" route from HTML page using AJAX call.

Output Screenshot:

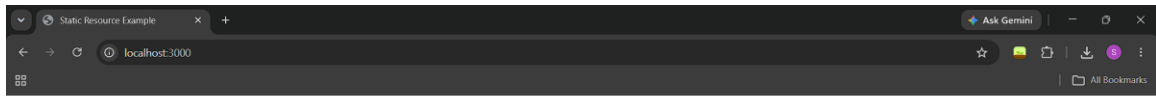


Problem 2

Problem Statement:

Develop a web server which serves static resources.

Output Screenshot:



Welcome to Node.js

This image is served as a static resource.

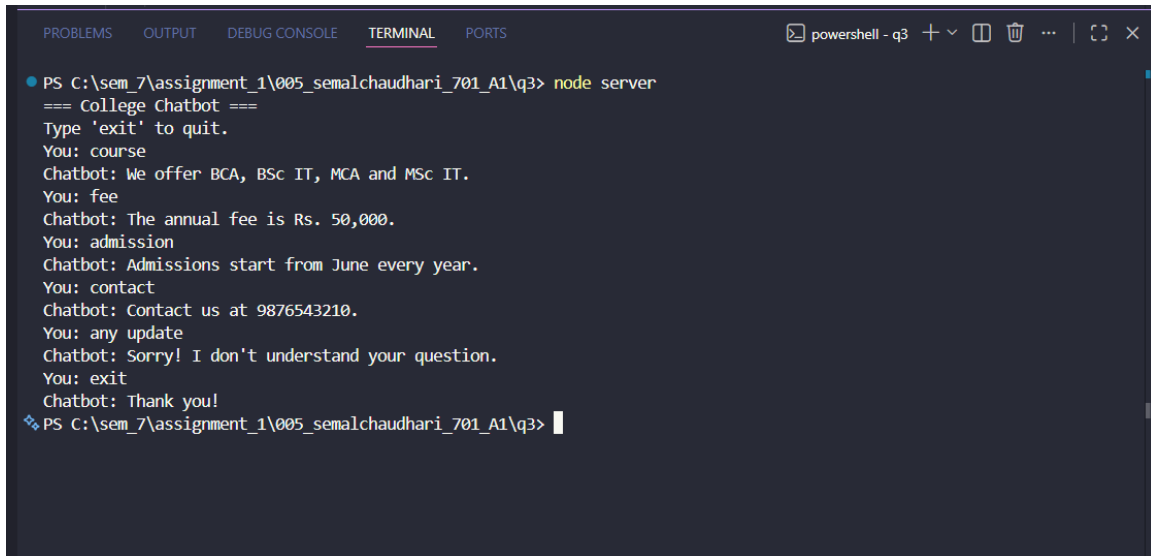


Problem 3

Problem Statement:

Develop a module for domain specific chatbot and use it in a command line application.

Output Screenshot:



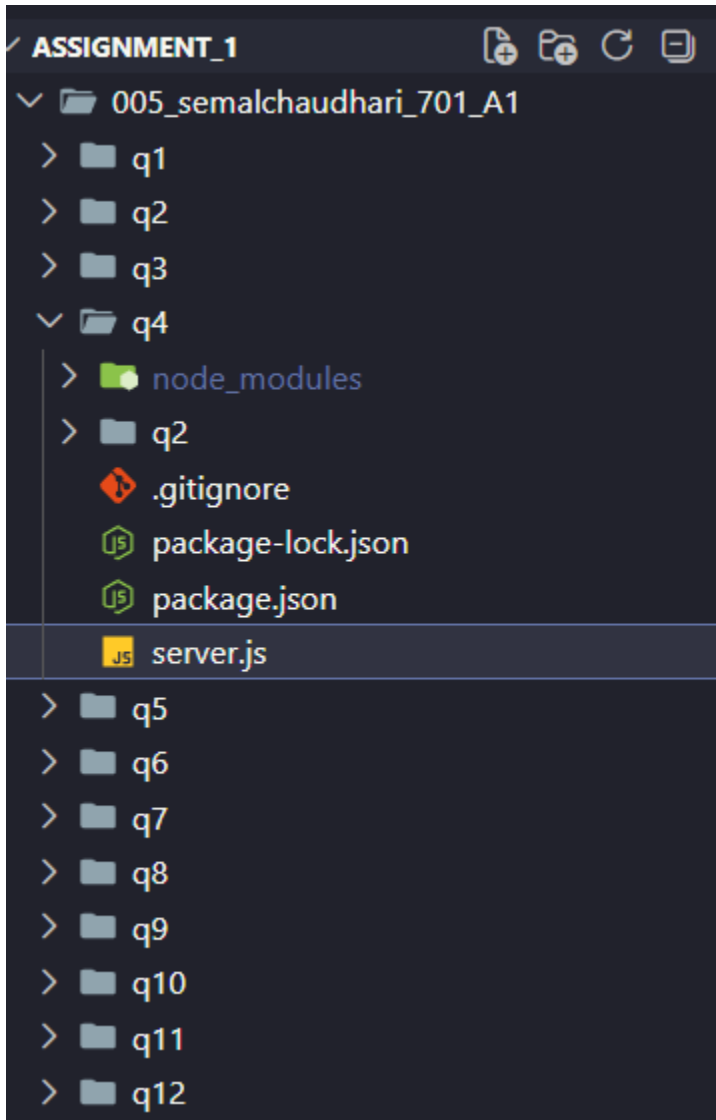
```
PS C:\sem_7\assignment_1\005_samalchaudhari_701_A1\q3> node server
=== College Chatbot ===
Type 'exit' to quit.
You: course
Chatbot: We offer BCA, BSc IT, MCA and MSc IT.
You: fee
Chatbot: The annual fee is Rs. 50,000.
You: admission
Chatbot: Admissions start from June every year.
You: contact
Chatbot: Contact us at 9876543210.
You: any update
Chatbot: Sorry! I don't understand your question.
You: exit
Chatbot: Thank you!
PS C:\sem_7\assignment_1\005_samalchaudhari_701_A1\q3>
```

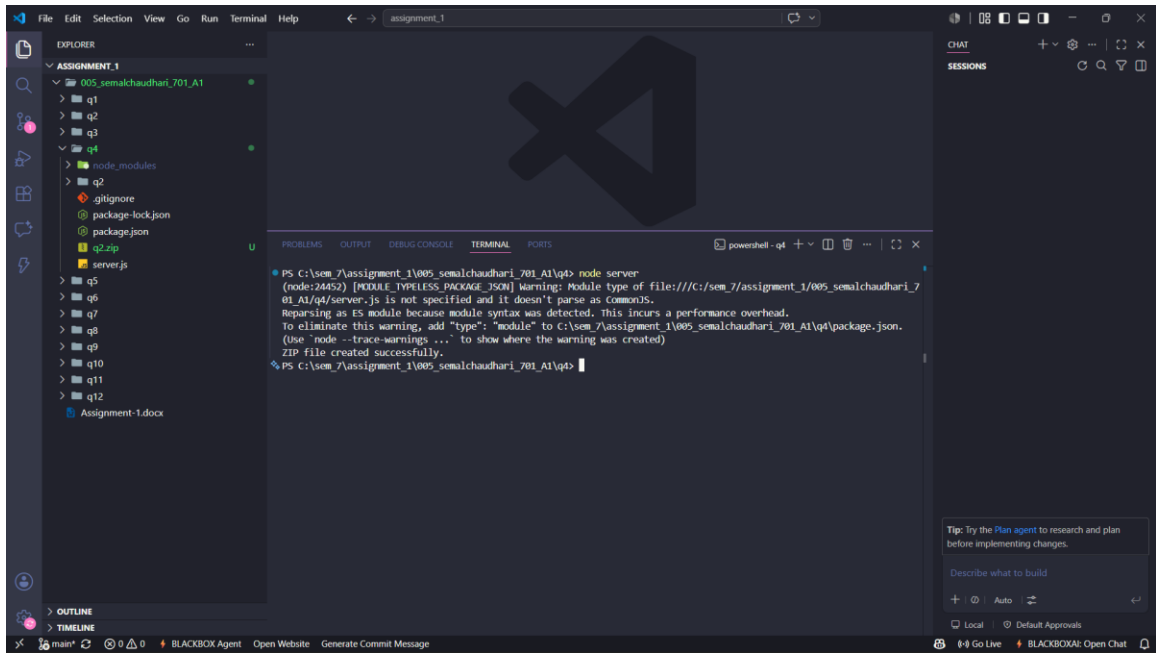
Problem 4

Problem Statement:

Write a program to create a compressed zip file for a folder.

Output Screenshot:



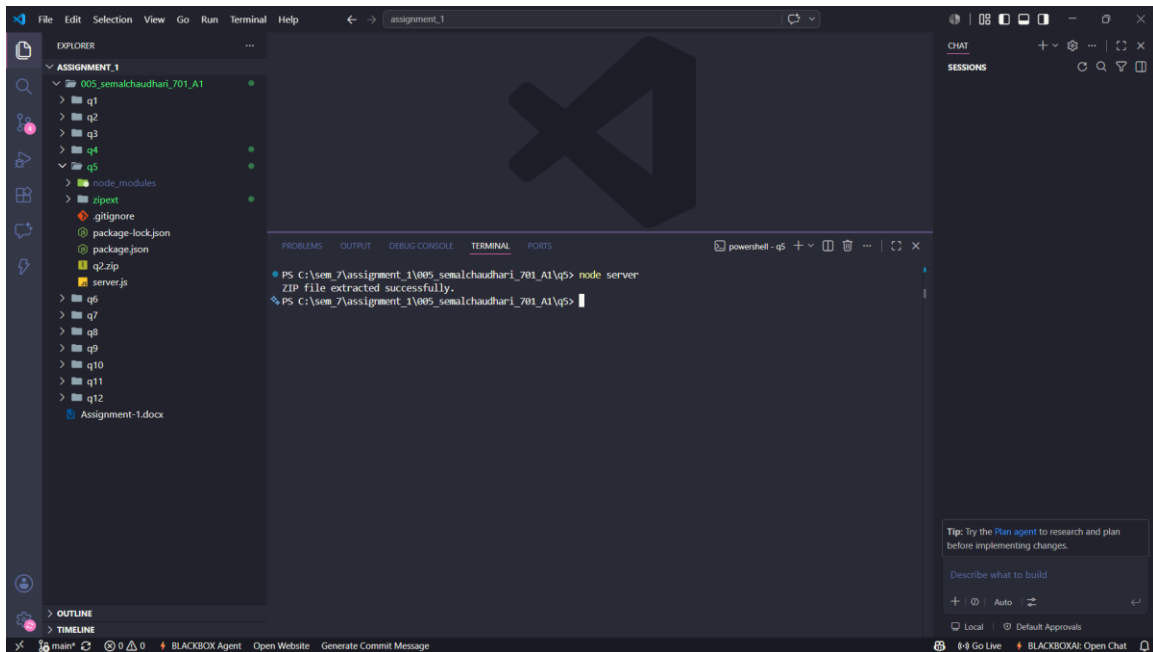
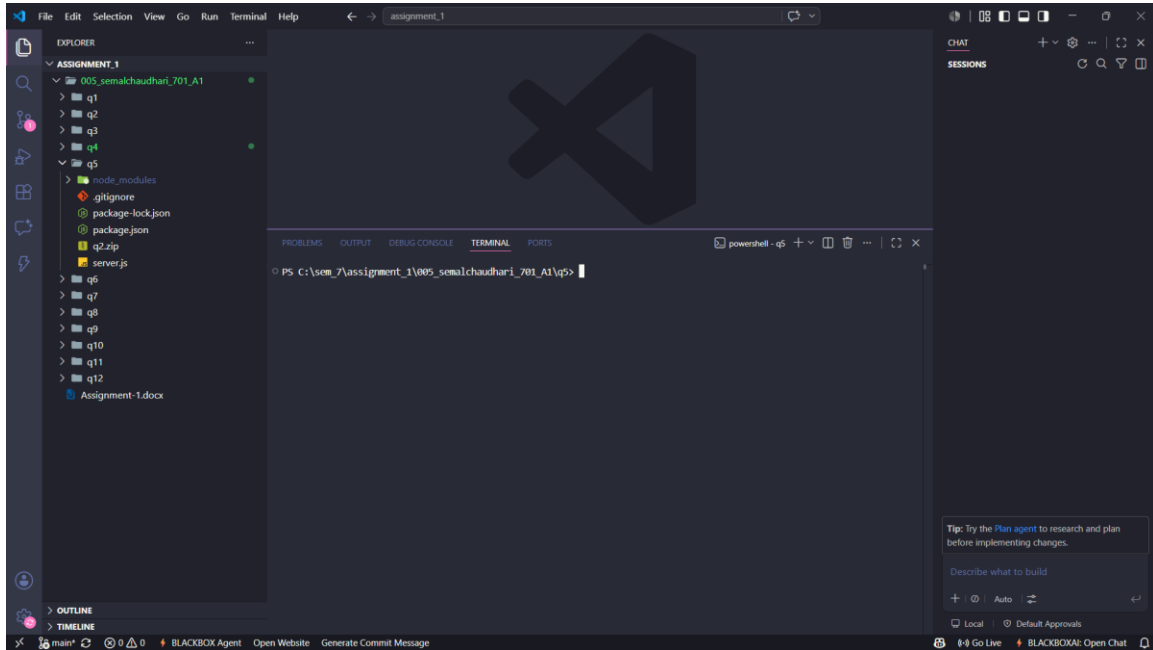


Problem 5

Problem Statement:

Write a program to extract a zip file.

Output Screenshot:

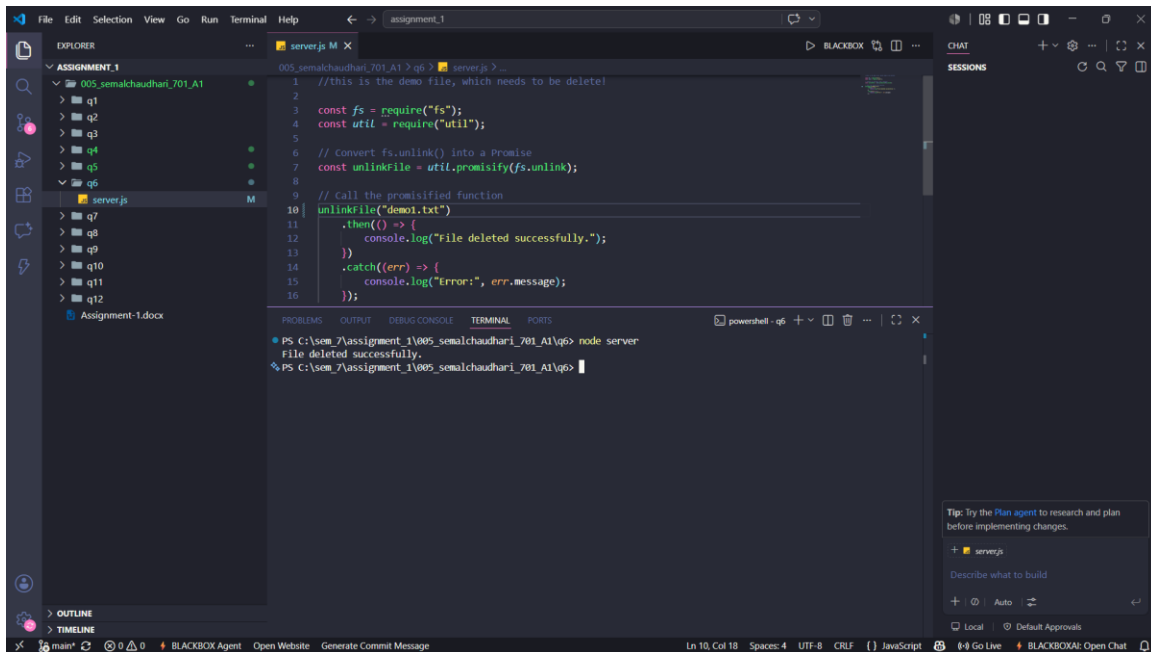
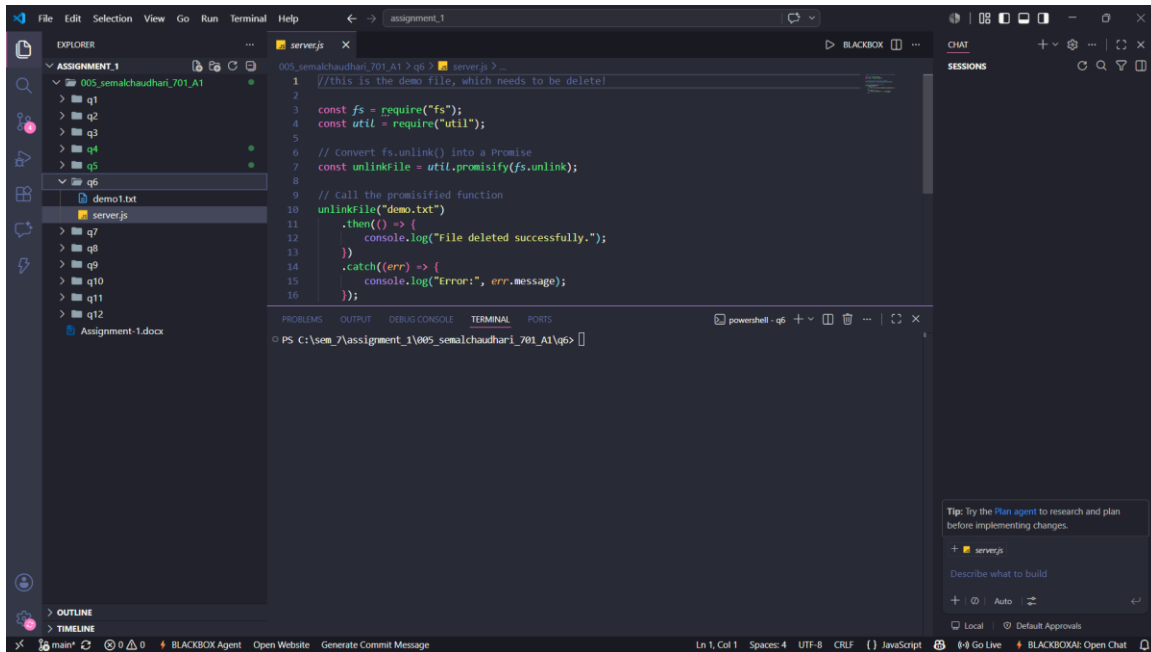


Problem 6

Problem Statement:

Write a program to promisify fs.unlink function and call it.

Output Screenshot:

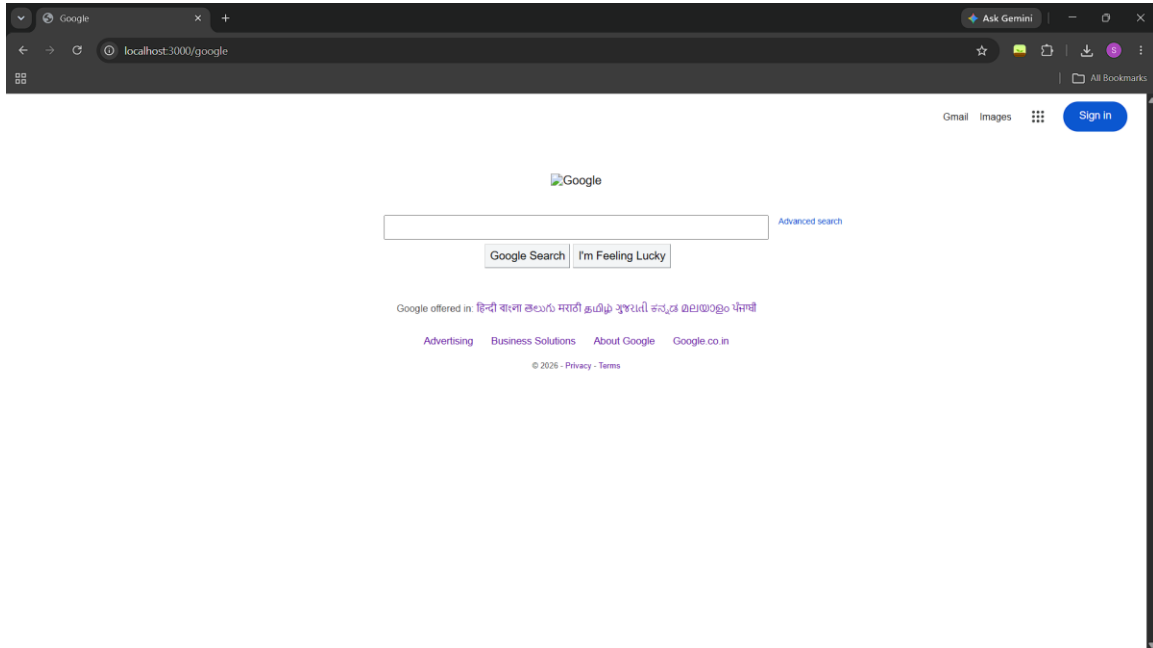


Problem 7

Problem Statement:

Develop a node web server with a route /google which can fetch data of google page using fetch() using async-await model and show as a response.

Output Screenshot:

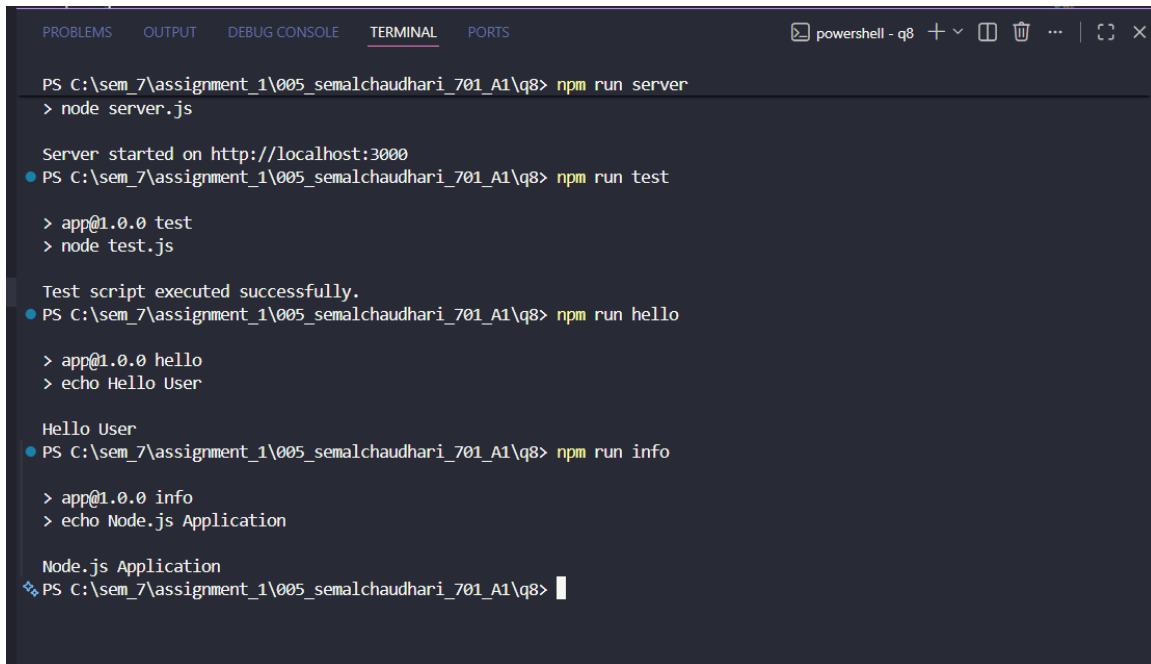


Problem 8

Problem Statement:

Set a server script, a test script and 3 user defined scripts in package.json file in your nodejs application.

Output Screenshot:



```
PS C:\sem_7\assignment_1\005_samalchaudhari_701_A1\q8> npm run server
> node server.js

Server started on http://localhost:3000
● PS C:\sem_7\assignment_1\005_samalchaudhari_701_A1\q8> npm run test

> app@1.0.0 test
> node test.js

Test script executed successfully.
● PS C:\sem_7\assignment_1\005_samalchaudhari_701_A1\q8> npm run hello

> app@1.0.0 hello
> echo Hello User

Hello User
● PS C:\sem_7\assignment_1\005_samalchaudhari_701_A1\q8> npm run info

> app@1.0.0 info
> echo Node.js Application

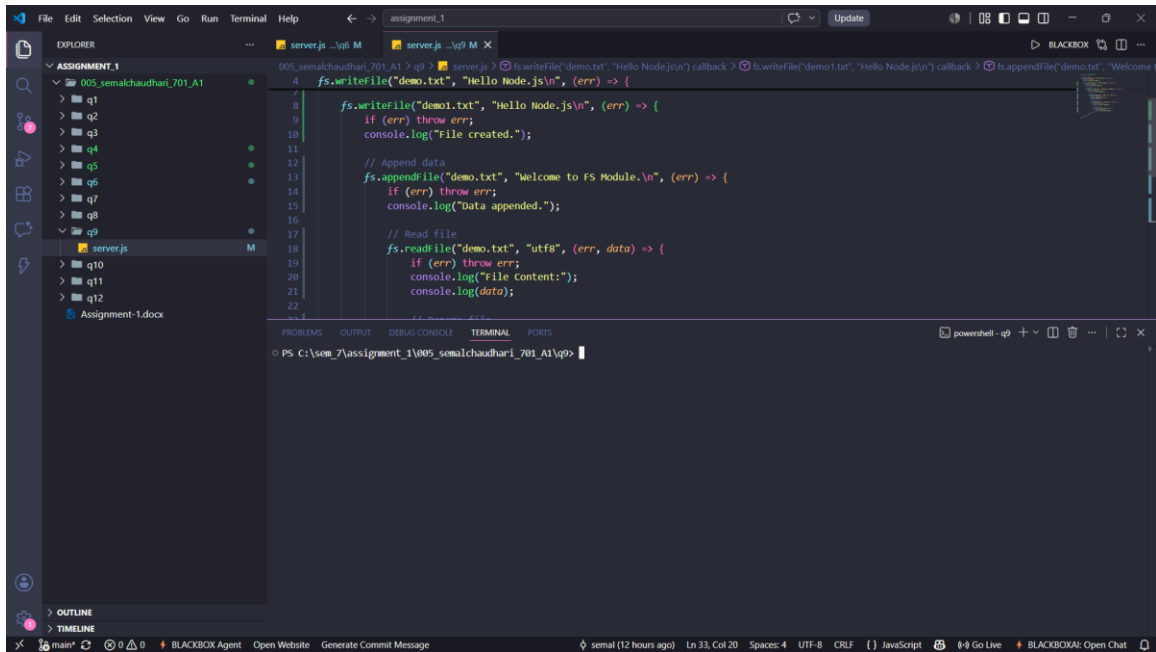
Node.js Application
❖ PS C:\sem_7\assignment_1\005_samalchaudhari_701_A1\q8>
```

Problem 9

Problem Statement:

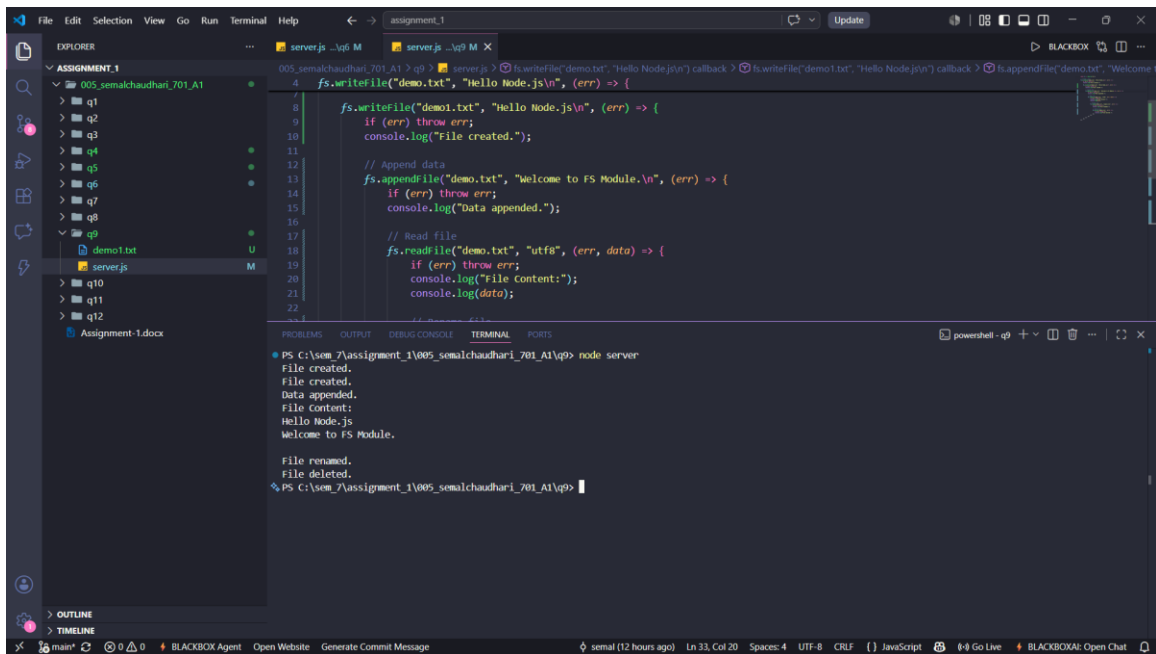
A program which calls useful functions in fs module.

Output Screenshot:



The screenshot shows the VS Code editor with a file named `server.js` open. The code uses the `fs` module to create a file, append data, and read the file content. The Explorer sidebar shows a project structure with a folder `005_samalchaudhari_701_A1` containing files `q1` through `q12`, and a file `server.js`. The terminal is currently empty.

```
1  const fs = require('fs');
2
3  // Create file
4  fs.writeFile('demo.txt', 'Hello Node.js\n', (err) => {
5
6  })
7
8  // Create file
9  fs.writeFile('demo1.txt', 'Hello Node.js\n', (err) => {
10
11 })
12
13 // Append data
14 fs.appendFile('demo.txt', 'Welcome to FS Module.\n', (err) => {
15
16 })
17
18 // Read file
19 fs.readFile('demo.txt', 'utf8', (err, data) => {
20
21 })
22
```



The screenshot shows the same VS Code editor with the `server.js` file open. The terminal now displays the output of the program after running the command `node server`. The output shows the file creation, data appending, and file reading process.

```
PS C:\sem_7\assignment_1\005_samalchaudhari_701_A1\q9> node server
File created.
File created.
Data appended.
File Content:
Hello Node.js
Welcome to FS Module.

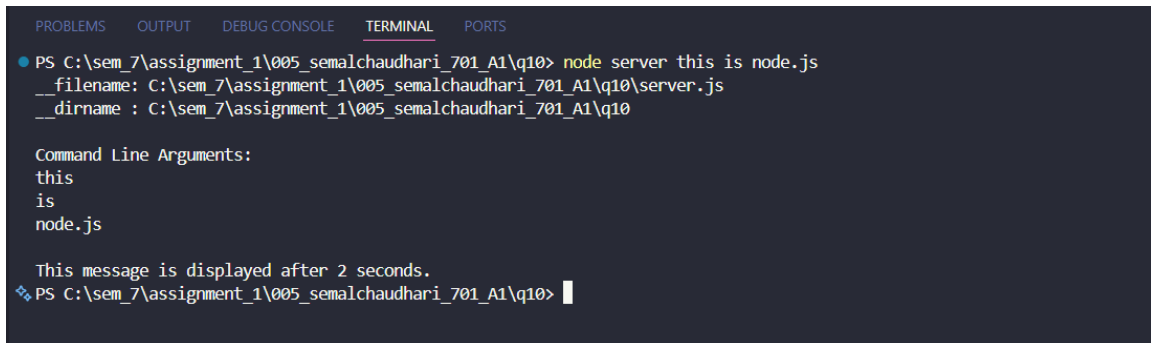
File removed.
File deleted.
PS C:\sem_7\assignment_1\005_samalchaudhari_701_A1\q9>
```

Problem 10

Problem Statement:

A program which uses global objects in nodejs. Also read command line arguments and print it.

Output Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS C:\sem_7\assignment_1\005_samalchaudhari_701_A1\q10> node server this is node.js
  __filename: C:\sem_7\assignment_1\005_samalchaudhari_701_A1\q10\server.js
  __dirname : C:\sem_7\assignment_1\005_samalchaudhari_701_A1\q10

Command Line Arguments:
this
is
node.js

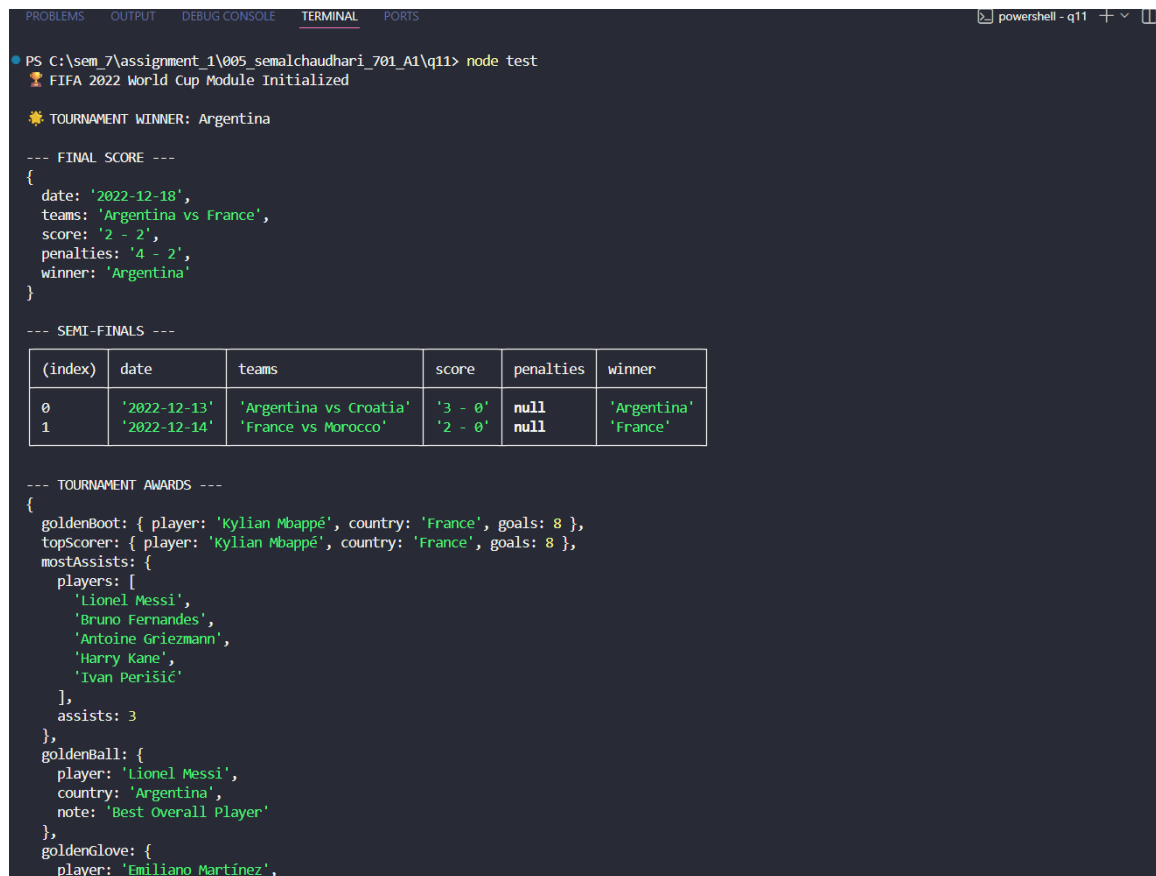
This message is displayed after 2 seconds.
❖ PS C:\sem_7\assignment_1\005_samalchaudhari_701_A1\q10> |
```

Problem 11

Problem Statement:

Develop a useful package and publish it on npmjs.com. Write a program to use it.

Output Screenshot:



```
PS C:\sem 7\assignment_1\005_samalchaudhari_701_A1\q11> node test
FIFA 2022 World Cup Module Initialized

TOURNAMENT WINNER: Argentina

--- FINAL SCORE ---
{
  date: '2022-12-18',
  teams: 'Argentina vs France',
  score: '2 - 2',
  penalties: '4 - 2',
  winner: 'Argentina'
}

--- SEMI-FINALS ---


| (index) | date         | teams                  | score   | penalties | winner      |
|---------|--------------|------------------------|---------|-----------|-------------|
| 0       | '2022-12-13' | 'Argentina vs Croatia' | '3 - 0' | null      | 'Argentina' |
| 1       | '2022-12-14' | 'France vs Morocco'    | '2 - 0' | null      | 'France'    |



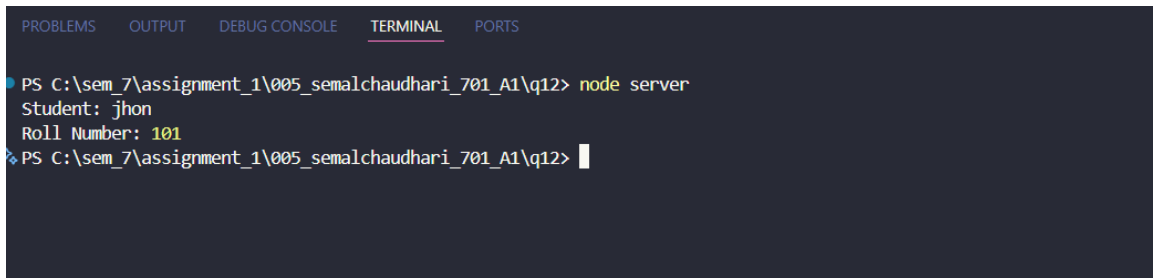
--- TOURNAMENT AWARDS ---
{
  goldenBoot: { player: 'Kylian Mbappé', country: 'France', goals: 8 },
  topScorer: { player: 'Kylian Mbappé', country: 'France', goals: 8 },
  mostAssists: {
    players: [
      'Lionel Messi',
      'Bruno Fernandes',
      'Antoine Griezmann',
      'Harry Kane',
      'Ivan Perišić'
    ],
    assists: 3
  },
  goldenBall: {
    player: 'Lionel Messi',
    country: 'Argentina',
    note: 'Best Overall Player'
  },
  goldenGlove: {
    player: 'Emiliano Martínez',
  }
}
```

Problem 12

Problem Statement:

Write a node program having at least 3 dependent tasks and call using chained promises.

Output Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\sem_7\assignment_1\005_samalchaudhari_701_A1\q12> node server
Student: jhon
Roll Number: 101
PS C:\sem_7\assignment_1\005_samalchaudhari_701_A1\q12> 
```